

SMART IRRIGATION USING SOIL-MOISTURE SENSING AND LoRa

Review 1 — Problem Statement and Literature Survey

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1. Problem statement

Irrigation on small farms is usually scheduled by time rather than by the actual water need of the crop. This results in over-watering after rainfall and under-watering during dry spells. Soil-moisture based control addresses this by switching the pump only when the volumetric water content at root depth falls below a crop-specific threshold.

2. Objectives

- Measure soil moisture at root depth across a 1-acre plot.
- Transmit readings to a gateway up to 2 km away with LoRa.
- Control the pump automatically and log every cycle.
- Provide a mobile dashboard with alerts and manual override.

3. Literature survey

#	Paper	Method	Limitation
1	Kumar et al., 2021	GSM-based pump control	SMS cost; poor coverage in fields
2	Rao & Iyer, 2022	Wi-Fi sensor mesh	Range under 100 m
3	Silva et al., 2023	LoRaWAN field study	Needs licensed network server
4	Chen et al., 2020	Resistive soil probes	Probe corrosion within months
5	Patel et al., 2022	ML-based irrigation schedule	Needs weather API connectivity

4. Research gap

Existing systems either rely on cellular connectivity or short-range radios. A low-cost, long-range, battery-powered design suitable for small farms is not well covered.